


Let K be a field

Def 1: Let $f(x) = a_n x^n + \dots + a_1 x + a_0 \neq 0$

be in $K[x]$. We say the leading term

of $f(x)$ is $LT(f(x)) = a_m x^m$ where

$m = \max \{d : a_d \neq 0\}$. We

then call the coefficient of the leading

term of $f(x)$ as the leading coefficient.

which we denote by $LC(f(x))$.

Lemma 2: Let $f(x), g(x) \in K[x]$.

Then. $LT(f(x)g(x)) = LT(f(x)) \cdot LT(g(x))$

and $LC(f(x)g(x)) = LC(f(x)) \cdot LC(g(x))$

Cor 3 : Let k be a field.

Then $k[x]$ is a domain.

proof. Suppose $0 \neq f(x)$ is a zero-divisor.

Then $\exists 0 \neq g(x) \in k[x]$ such that

$$f(x)g(x) = 0 \text{ which implies}$$

$$LC(f(x)) \cdot LC(g(x)) = 0 \text{ in } k.$$

It follows $LC(f(x)) = 0$ or $LC(g(x)) = 0$.

Cor 4: let k be a field. Then.

$0 \neq f(x) \in k[x]$ is a unit iff $\deg f = 0$.

proof. We note that

$0 \neq f(x) \in k[x]$ is a unit

iff $\exists 0 \neq g(x) \in k[x]$ s.t

$$f(x)g(x) = 1$$

iff $\deg f(x) + \deg g(x) = 0$

iff $\deg f(x) = \deg g(x) = 0$

defn. Let $f(x), g(x) \in k[x]$.

We say $f(x)$ divides $g(x)$ if

$$\exists h(x) \in k[x]. \text{ s.t. } h(x)f(x) = g(x)$$

We say $f(x) \in k[x]$ is irreducible

if $f(x) \neq 0$ is not a unit

and whenever $f(x) = g(x)h(x)$

for $g(x), h(x) \in k[x]$. $\deg g(x) = 0$

or $\deg h(x) = 0$.

eg. $x^2 + 1$ is irreducible in $\mathbb{R}[x]$.

factors as $(x+i)(x-i)$ in $\mathbb{C}[x]$

Def 6. Let R, S be commutative rings

A function $f: R \rightarrow S$ is a ring homomorphism

if $f(1_R) = 1_S$

$$f(a+b) = f(a) + f(b)$$

$$f(ab) = f(a)f(b) \quad \forall a, b \in R.$$

eg. Let K be a field and $p \in K$.

Define $ev_p: K[x] \rightarrow K$

$$\sum_{i=0}^n a_i x^i \mapsto f(x) \mapsto f(p) = \sum_{i=0}^n a_i p^i$$

This is a ring homomorphism called

evaluation at p .

def 7: let $p \in k$. We say p is a

root of $f(x) = \sum_{i=0}^n a_i x^i$ if

$$ev_p(f) = f(p) = 0 \text{ in } k.$$

Theorem 8. Let k be a field.

Then $k[x]$ is an Euclidean domain.

with respect to the degree function.

proof. We show $\forall f(x), g(x) \in k[x]$

for $g(x) \neq 0$. we may write

$$f(x) = g(x) q(x) + r(x)$$

where $q(x), r(x) \in k[x]$ and

$$r(x) = 0 \quad \text{or} \quad \deg r(x) < \deg g(x).$$

We prove by (strong) induction on the degree of $f(x)$.

If $\deg f(x) = 0$, then this is clearly true regardless of $\deg g(x)$.

Suppose the claim is true for all

$$f(x) \in k[x] \quad \text{of} \quad \deg f(x) < n.$$

Let $f(x)$ be of degree n .

$$\text{If} \quad \deg g(x) > \deg f(x).$$

$$\text{then let } q(x) = 0, \quad r(x) = f(x).$$

if $\deg g(x) \leq \deg f(x)$.

$$\text{let } q'(x) = \frac{LT f(x)}{LT g(x)} \in k[x].$$

and note $f(x) - q'(x)g(x) = f'(x)$

is of $\deg < n$.

By induction hypothesis.

$$f'(x) = g(x)q''(x) + r(x)$$

for $q''(x), r(x) \in k[x]$.

and $r(x) = 0$ or $\deg r(x) < \deg g(x)$

$$\begin{aligned} \text{It follows } f(x) &= (q'(x) + q''(x)g(x))g(x) + r(x) \\ &= q(x)g(x) + r(x). \end{aligned}$$

Cor 9. Let $p \in K$ and $f(x) \in K[x]$.

Then. p is a root of $f(x)$

iff $x-p \mid f(x)$.

proof: ($\Leftarrow$) If $f(x) = (x-p)g(x)$

for some $g(x) \in K[x]$, then

$$\text{ev}_p(f) = \text{ev}_p(x-p) \text{ev}_p(g) = 0.$$

($\Rightarrow$) Write $f(x) = (x-p)g(x) + r(x)$

where $r(x) = 0$ or $\deg r(x) < \deg x-p = 1$

In particular, $r(x)$ is a constant polynomial.

Since p is a root, $\text{ev}_p(f) = \text{ev}_p(x-p) \text{ev}_p(g) + \text{ev}_p r$

so $0 = \text{ev}_p(f) = \text{ev}_p(r) = r$. Then $f(x) = (x-p)g(x)$.